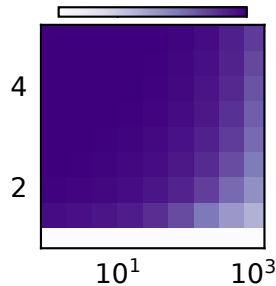


total harm  
0.5 1.0

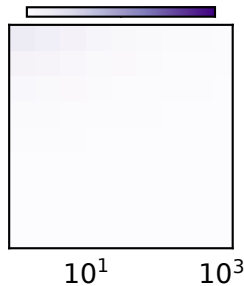
reproduction  
number,  $R_0$



infection  
0.5 1.0

4

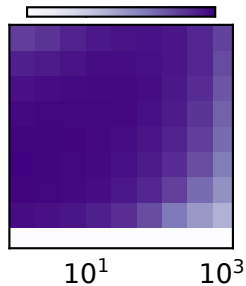
2



toxicity  
0.5 1.0

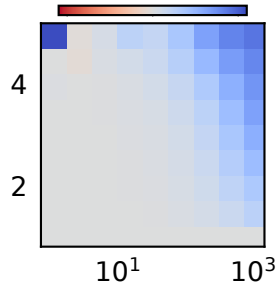
4

2



$\psi_{Ag}$   
-2.5 0.0 2.5

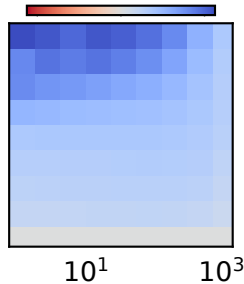
reproduction  
number,  $R_0$



$\psi_{inf}$   
-2 0 2

4

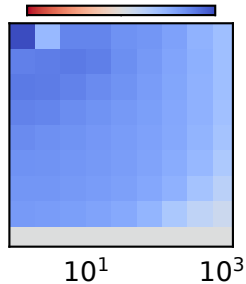
2



$\psi_{res}$   
-2.5 0.0 2.5

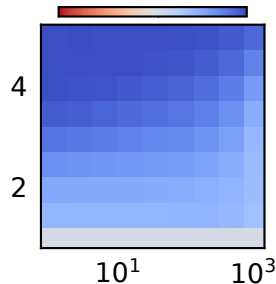
4

2



$g_{0,N \rightarrow N^*} = g_{0,E \rightarrow 2E}$   
0.0 0.5

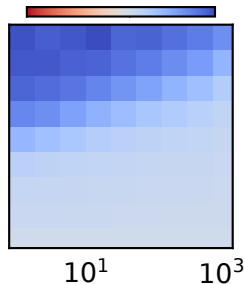
reproduction  
number,  $R_0$



$g_{0,N^* \rightarrow E}$   
0.0 0.5

4

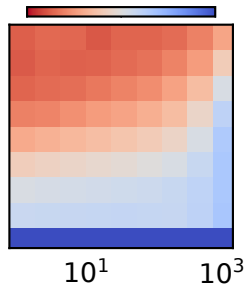
2



$g_{0,E \rightarrow \emptyset}$   
0.00 0.25 0.50

4

2



immunogenicity,  $\kappa$

immunogenicity,  $\kappa$

immunogenicity,  $\kappa$